NATALIA BURGIEL

HM # 7

26.5-2

$HETP = N_t \cdot H$

↑ theoretical minimum # of trays

↑ hight of packing

$E_o = \frac{\text{\# of theoretical minimum trays}}{\text{\# of actual trays}}$

$\rho_L = 57.9 \ lb_m/ft^3 = 927 \ kg/m^3$

$\mu_L = 1.4 cp$ (viscosity)

$M_L = 245$

$y = mx = 0.7x \Rightarrow m = 0.7$

$$E_o = \frac{\log\left[1 + E_m\left(m\frac{V}{L} - 1\right)\right]}{\log\left(m\frac{V}{L}\right)}$$

$$\log E_o = 1.597 - 0.199 \log\left(\frac{m M_L \mu_L}{\rho_L}\right) - 0.0896\left[\log\left(\frac{m M_L \mu_L}{\rho_L^2}\right)\right]^2$$

$$\log E_o = 1.597 - 0.199 \log\left(\frac{0.7 \cdot 245 \cdot 1.4}{57.9}\right) - 0.0896\left[\log\left(\frac{0.7 \cdot 245 \cdot 1.4}{57.9}\right)\right]^2 =$$

$= 1.4399$

$E_o = 27.54 \%$

For Norton Intelox 2T (From Table 22.3-1)

$a = 65 \ ft^2/ft^3 = 213 \ m^2/m^3$

$$HETP = \frac{100}{a} + 0.33 = \frac{100}{65} + 0.33 = 1.868 \ ft$$

~~$HETP = \frac{100}{a} + 0.1 = \frac{100}{213} + 0.1 = 0.57 \ m$~~

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26.6-1

$\rho_L = 673\ kg/m^3$

$\rho_V = 3.68\ kg/m^3$

$\sigma = 22.5\ dyn/cm = 0.0225$

$V = 21000\ kg/hr$

$L = 19500\ kg/hr$

$$\frac{L}{V}\left(\frac{\rho_V}{\rho_L}\right)^{0.5} = \frac{19500}{21000}\left(\frac{3.63}{673}\right)^{0.5} = 0.0682$$

From Figure 26.6-1 in the textbook

for 24 in. spacing

$0.37 \cdot 0.95 = K_V = 0.35\ ft/s = 0.1066813$

foaming factor

$$V_{max} = K_V\left(\frac{\sigma}{20}\right)^{0.2}\sqrt{\frac{\rho_L-\rho_V}{\rho_V}} = 0.1068\left(\frac{22.5}{20}\right)^{0.2}\sqrt{\frac{673-3.68}{3.68}} = 1.473\ m/s$$

(downspout area + 80% flooding) $1.473 \cdot 0.8 \cdot 0.91 = 1.072\ m/s$

$$\frac{21000}{3600}\cdot\frac{1}{3.68}\cdot\frac{1}{1.072} = 1.4785\ m^2$$

$$\frac{\pi D^2}{4} = 1.478 \Rightarrow D = \sqrt{\frac{1.478\cdot 4}{\pi}} = 1.372\ m$$

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26.7-4

$x_D = 0.98$

$x_W = 0.04$

$R = 1.3 R_m = 1.3 \cdot 1.17 = 1.521$

$F = 100$

$x_F = 0.55$ @ boiling point liquid $\rightarrow q = 1$

$\frac{L}{D} = 1.521$

$F = 100 = D + W$

$0.55 \cdot 100 = 0.98 D + 0.04 W$

$55 = 0.98 \cdot (100 - W) + 0.04 W$

$55 = 98 - 0.98 W + 0.04 W$

$-43 = -0.94 W$

$45.745 = W$

$D = 100 - 45.745 = 54.255$

$L = 1.521 \cdot 54.255 = 82.513$

$V_1 = L + D = 82.513 + 54.255 = 136.777$

$y_A = x_D$

$H_1 = y_A [\lambda_A + c_{pyA}(T - T_0)] + (1 - y_A)[\lambda_B + c_{pyB}(T - T_0)] =$

$= 0.98[30820 + 96.3(82.3 - 80.1)] + (1 - 0.98)[34224 + 138.2(82.3 - 80.1)] =$

$= 31101.78$

$h_D = 0.98(138.2)(81.1 - 80.1) + (1 - 0.98)(167.5)(81.1 - 80.1) = 138.79$

$y_{n+1} = \frac{L_n}{V_{n+1}} x_n + \frac{D x_D}{V_{n+1}} = \frac{82.513}{136.777} x_n + \frac{54.255 \cdot 0.98}{136.777} = 0.603 x_n + 0.389 =$

$q_c = V_1 H_1 - L h_D - D h_D$

$q_c = 136.777 \cdot 31101.78 - 82.513 \cdot 138.79 - 54.255 \cdot 138.79 =$

$\boxed{q_c = 4235026.132 \text{ kJ/h} \quad \text{condenser heat load}}$

Based on the Excel calculations theoretical # of trays is 11.5.

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$q_R = D h_D + W h_W + q_c - F h_f$

Using Fig 26.7-1 for $x_W = 0.04$ $h_W = 4420$

$h_f = 0.55(132.2)(54.5 - 80.1) + (1 - 0.55)(167.5)(54.5 - 80.1) =$

$= ~~-4945.856~~$

$= -3875.456$

$q_R = (54.255 \cdot 138.79) + (45.75 \cdot 4420) + 342350 26.132 - 100 \cdot (-3875.456)$

$= 4047161.18$ kJ/h

Minimal # of trays is 9.7

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26.8-4

26.8-1

Used Fig. 26.8-2 for K- values

a)

Trial I (assume 106°C)

Comp	y_{iD}	K_i	$\alpha_i = \frac{K_i}{K_c}$	y_i/α_i
A	0.4	3.5	5.036	0.0794
B	0.25	1.54	2.216	0.1128
C	0.2	0.695	1.000	0.2000
D	0.15	0.33	0.4748	0.3159
	1.00			$\Sigma y_i/\alpha_i = 0.7081$

0.7081 > 0.695 at 106°C

Trial II (assume 107°C)

Comp	y_{iD}	K_i	$\alpha_i \frac{K_i}{K_c}$	y_i/α_i	$\frac{y_i/\alpha_i}{\Sigma y_i/\alpha_i} = x_i$
A	0.4	3.55	4.931	0.0811	0.114 = x_A
B	0.25	1.6	2.222	0.1125	0.158 = x_B
C	0.2	0.72	1.000	0.2000	0.281 = x_C
D	0.15	0.34	0.472	0.3178	0.447 = x_D
	1.00			$\Sigma y_i/\alpha_i = 0.7114$	1.000

liquid composition

$K_{C_{calc}} = \Sigma y_i/\alpha_i = 0.7114$

close to $K_{C_{measured}} = 0.720$ $\boxed{T_{dew} = 107°C}$

b) Partial, $f = 0.4$

Trial I (assume 85°C)

Comp	x_{iF}	K_i	α_i	$K_c\alpha_i - 1$	$f(K_c\alpha_i - 1) + 1$	$\frac{x_iF}{f(K_c\alpha_i-1)+1}$
A	0.4	2.45	5.698	1.450	1.580	0.2532
B	0.25	1.01	2.349	0.010	1.004	0.2490
C	0.2	0.43	1.000	-0.570	0.772	0.2591
D	0.15	0.185	0.430	-0.815	0.674	0.2256
	1.00					$\Sigma x_i = 0.9869$

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Trial II

(assume 82°)

Comp	x_{iF}	K_i	α_i	$K_c\alpha_i - 1$	$f(K_c\alpha_i - 1) + 1$	$\frac{x_{iF}}{f(K_c\alpha_i - 1)+1}$
A	0.40	2.34	5.85	1.34	1.536	0.2604
B	0.25	0.95	2.375	-0.05	0.980	0.2550
C	0.20	0.40	1.00	-0.6	0.760	0.2632
D	0.15	0.17	0.425	-0.83	0.668	0.2246
						$\Sigma x_i = 1.0032$

So, T = 82°C

$x_A = 0.26$
$x_B = 0.254$
$x_C = 0.262$
$x_D = 0.224$
1.000

Liquid

F = 100 mol

Comp	x_{iF}	$x_{iF}F$	$x_iL = L_i$	$x_iF - L_i = V_i$
A	0.4	40.0	0.260 · 60 = 15.6	24.4
B	0.25	25.0	0.254 · 60 = 15.24	9.76
C	0.2	20.0	0.262 · 60 = 15.72	4.28
D	0.15	15.0	0.224 · 60 = 13.44	1.56
		100.0	60.0	40.0

$V_i/40 = y_i$

$y_A = 0.61$
$y_B = 0.244$
$y_C = 0.107$
$y_D = 0.039$
1.000

Vapor

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26.8-4

$F = 100$ mol/h

x_{AF} (nC_4) = 0.4

x_{BF} (nC_5) = 0.25

x_{CF} (nC_6) = 0.2

x_{DF} (nC_7) = 0.15

a) Material Balance

Comp. B. $x_{BF} F = 0.25 \cdot 100 = y_{BD} D + x_{BW} W$

$y_{BD} D = 0.95 \cdot 25 = 23.75$ (95% B)

$x_{BW} W = 0.05 \cdot 25 = 1.25$

Comp. C. $x_{CF} F = 0.2 \cdot 100 = 20 = y_{CD} D + y_{CW} W$

$x_{CW} W = 0.95 \cdot 20 = 19$

$y_{CD} D = 0.05 \cdot 20 = 1.0$

Assume no comp. D in distillate and no A in W for I trial.

Therefore, $y_{AD} D = 0.4 \cdot 100 = 40$

$x_{DW} W = 0.15 \cdot 100 = 15$

Comp	Feed F x_F	$x_F F$	D, Distillate $y_D D$	$y_D = x_D$	Bottoms, W $x_W W$	x_W
A	0.4	40	40	0.6178	0	0
B (L)	0.25	25	23.75	0.3668	1.25	0.0355
C (H)	0.20	20	1.00	0.0154	19.00	0.5390
D	0.15	15	0.00	0.00	15.00	0.4255
	1.000	100	64.75	1.000	W = 35.25	1.000

b) Dew point

Trial I

Assume 67°C

Comp	y_{iD}	K_i	α_i	y_i/α_i
A	0.6178	1.75	6.73	0.0918
B (L)	0.3668	0.65	2.50	0.1467
C (H)	0.0154	0.26	1.00	0.0154
D	0	0.10	0.385	0

$K_{c\,calc} = \sum y_i/\alpha_i = 0.2539$

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Comp	K_i	α_i	y_i/α_i
A	1.72	6.745	0.0916
B	0.645	2.529	0.145
C	0.255	1.00	0.0154
D	0.095	0.373	0

Trial II

Assume 66°C

$K_{c\,calc} = \Sigma y_i/\alpha_i = 0.2520$

$$\boxed{\begin{aligned} x_i &= \frac{y_i/\alpha_i}{\Sigma y_i \alpha_i} \\ x_A &= 0.3635 \\ x_B &= 0.5754 \\ x_C &= 0.0611 \\ x_D &= 0 \\ &\overline{\quad 1.000 \quad} \end{aligned}}$$

$T_{dew} = 66°C$

Boiling point

Comp	X_{iw}	K_i	α_i	$\alpha_i X_i$
A	0	5.1	4.322	0
B	0.0355	2.4	2.034	0.0722
C	0.5390	1.18	1.000	0.5390
D	0.4255	0.63	0.534	0.2272
	1.000			

$\Sigma \alpha_i x_i = 0.8384$

Trial I

Assume 133°C

Comp	K_i	α_i	$\alpha_i X_i$
A	5.2	4.370	0
B	2.45	2.059	0.0731
C	1.19	1.000	0.5390
D	0.64	0.538	0.2288

$\Sigma \alpha_i x_i = 0.8409$

Trial II

Assume 134°C

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$$\frac{\alpha_i x_i}{\Sigma \alpha_i x_i} = y_i \qquad T_{boil} = 134°C$$

$y_A = 0$
$y_B = 0.0869$
$y_C = 0.6410$
$y_D = 0.2721$

1.000

$$K_{c\,calc} = \frac{1}{\Sigma \alpha_i x_i} = 1.193\ (134°C) = 1.189$$

c) $\alpha_{LD} = 2.529\ (66°C)$

$\alpha_{LW} = 2.059\ (134°C)$

$$\alpha_{L,avg} = \sqrt{2.529 \cdot 2.059} = 2.282$$

$$N_m = \frac{\log[(x_{LD}D / x_{HD}D)(x_{HW}W / x_{LW}W)]}{\log(\alpha_{L,avg})} =$$

$$= \frac{\log[(0.3668 \cdot 64.75 / 0.0154 \cdot 64.75)(0.539 \cdot 35.25 / 0.0355 \cdot 35.25)]}{\log(2.282)} =$$

$$= \boxed{7.14 \text{ theoretical stages}}$$

Comp	α_D	α_W	$\sqrt{\alpha_D \alpha_W}$
A	6.745	4.370	5.429
B (L)	2.529	2.059	2.282
C (H)	1.000	1.000	1.000
D	0.373	0.538	0.448

A

$$\frac{x_{AD}D}{x_{AW}W} = (\alpha_{A,avg})^{N_m} \frac{x_{HD}D}{x_{HW}W} = (5.429)^{7.14}\left(\frac{0.0154 \cdot 64.75}{0.539 \cdot 35.25}\right) = 9229$$

Balance on A

$$x_{AF}F = 40 = x_{AD}D + x_{AW}W = 9229\, x_{AW}W + x_{AW}W$$

$x_{AW}W = 0.0043$ mol

$x_{AD}D = 34.996$ mol

D

$$\frac{x_{DD}D}{x_{DW}W} = (0.448)^{7.139}\left(\frac{0.0154 \cdot 64.75}{0.539 \cdot 35.25}\right) = 1.710 \cdot 10^{-4}$$

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Balance on D

$x_{DF} F = 15 = x_{DD} D + x_{DW} W = 1.710 \cdot 10^{-4} x_{DW} W + x_{DW} W$

$$\left| \begin{array}{l} x_{DW} W = 14.997 \\ x_{DD} D = 0.0026 \end{array} \right|$$

Comp	D: $y_D = x_D$	D: $x_D D$	W: x_W	W: $x_W W$
A	0.61776	39.996	0.00012	0.0043
B	0.3668	23.750	0.0355	1.250
C	0.0154	1.000	0.5390	19.000
D	0.00004	0.0026	0.4254	14.997
	1.000	64.75 = D	1.000	35.251 = W

Trace compositions : $x_{AW} = 1.2 \cdot 10^{-4}$ (traces too small)

$y_{DD} = 4 \cdot 10^{-5}$

d) $T_{avg} = \dfrac{(T_{top} + T_{bottom})}{2} = \dfrac{66+134}{2} = 100°C$

Comp	x_{iF}	K_i	α_i	x_{iD}
A	0.4	3.15	5.25	0.6177
B	0.25	1.38	2.3	0.3668
C	0.2	0.6	1.00	0.0154
D	0.15	0.285	0.475	0.0004
	1.000			1.000

$$1-q = 1-1 = 0 = \frac{5.25(0.4)}{5.25-\theta} + \frac{2.3(0.25)}{2.3-\theta} + \frac{1(0.2)}{1-\theta} + \frac{0.475(0.15)}{0.475-\theta}$$

Trial and error

θ	$\frac{2.10}{5.25-\theta}$ +	$\frac{0.575}{2.3-\theta}$ +	$\frac{0.2}{1-\theta}$ +	$\frac{0.0713}{0.475-\theta}$	$= \Sigma$
1.210	0.5198	0.5275	−0.9524	−0.0970	−0.0021
1.2096	0.5198	0.5273	−0.9542	−0.0971	−0.0042
1.2104	0.5199	0.5277	−0.9506	−0.0970	−0.0.0

Therefore $\theta = 1.2104$

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$$R_m + 1 = \frac{\sum \alpha_i x_{iD}}{\alpha_i - \theta} = \frac{5.25(0.6177)}{5.25 - 1.2104} + \frac{2.3(0.3668)}{2.3 - 1.2104} + \frac{1(0.0514)}{1 - 1.2104} + \frac{0.475(0.0004)}{0.475 - 1.2104}$$

$R_m + 1 = 1.5039$

$\boxed{R_m = 0.504}$

e) $R = 1.3(0.504) = 0.6552$

$\frac{R}{R+1} = \frac{0.6552}{1.6552} = 0.3958$

$\frac{R_m}{R_m+1} = \frac{0.504}{1.504} = 0.3351$

$\frac{N_m}{N} = 0.425$

$N = \frac{N_m}{0.425} = \frac{7.14}{0.425} = \boxed{16.8 \text{ theoretical stages}}$

f) $\log \frac{N_e}{N_s} = 0.206 \log \left[\frac{X_{HF}}{X_{LF}} \frac{W}{D} \left(\frac{X_{LW}}{X_{HD}} \right)^2 \right] = 0.206 \log \left[\frac{0.2}{0.25} \frac{35.251}{64.75} \left(\frac{0.0355}{0.0154} \right)^2 \right] =$

$= 0.075075$

$\frac{N_e}{N_s} = 1.188$

$N_e = 1.188 N_s$

$N = N_e + N_s = 16.8$

$1.188 N_s + N_s = 16.8$

$\boxed{N_s = 7.7 \text{ stages}}$
$\boxed{N_e = 9.1 \text{ stages}}$

Feed 9.1 stages from top

26.8-7

a)

$$F = D + W$$

$$100 = D + W$$

$$x_{AF} F = x_{AD} D + x_{AW} W$$

$$0.047 \cdot 100 = 0.126 D + 0W$$

$$4.7 = 0.126 D$$

$$37.3 \text{ mol/h} = D$$

$$W = 100 - D = 100 - 37.3 = 62.7 \text{ mol/h}$$

$$N_m = \frac{\log\left[\left(\frac{0.1913 \cdot 37.3 / 0.6827 \cdot 37.3}{0.23}\right)\left(0.999 \cdot 62.7 / 0.001 \quad 62.7\right)\right]}{\log(1.58)} =$$

$\alpha_A = 4.19$

$\alpha_B = 1.58$ $\qquad = 12.14$ steps

$\alpha_C = 1.00$

$$\frac{x_{AD} D}{x_{AW} W} = (\alpha_A)^{N_m} \cdot \frac{x_{HD} D}{x_{HW} W} =$$

$$= (4.19)^{12.14} \cdot \frac{0.6827 \cdot 37.3}{0.199 \quad 62.7} = 1454735 1.89$$

$$x_{AF} F = 0.047 \cdot 100 = x_{AD} D + x_{AW} W$$

$$4.7 = 14547352\, x_{AW} W + x_{AW} W$$

$$4.7 = x_{AW} W (14547372 + 1)$$

$$0.0000032 = x_{AW} W$$

$$47 = x_{AD} D$$

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